

Parcel Scanner Sim – Test Plan

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1. Introduction

Main purpose of this document is to create a test plan of my mobile application, first I want to show general functionality of it, create plan and test cases to practice my skills in QA field of mobile applications.

Base of application is built on android native application stack practiced and used by me in my projects, which is Android Studio as a main IDE, Kotlin as programming language paired with Jetpack Compose to build components and UI used across application, using Material Design 3 that declare main functionalities look and theme, standard and well-known navigation libraries used to navigate between screens, Retrofit and OkHttp for communication with API endpoints, and last two paired libraries that are MLKit (Text reader OCR and Barcode scanner) with declared algorithms and CameraX to use system camera without any troubles.

Application by itself is built in MVVM architecture that in my opinion is well designed and always finds appropriate way to use it in small-medium scale applications. As with every project I try to make, I want to learn something new, so in this application I implemented a skeleton UI architecture that is separation of Scaffold from displayed dynamic content, making changes in screens and displayed components like TopBar and BottomBar easier and more available.

2. Scope of testing

In Scope:

Result of our testing process should be ready to be published and used in portfolio applications, where all screens and main components work without problems. Main functionality and connection across all components are ready to be used.

Focus of test scope should be:

- NFC Logon System – Validation, edge cases, invalid characters
- Home – Screen UI display, components functionality, data validation
- History – Screen UI display, API connection validation, data validation, components functionality, limited system resources (network, device problems, interrupt system)
- Profile – Screen UI display, data validation, API connection validation, user session validation, edge cases, components functionality
- Package Scanner – functionality tests, edge cases, camera integration, components functionality, error handling, negative testing
- Form Scanner – functionality tests, edge cases, OCR functionality, edge cases, negative testing, error handling

Out Of Testing Scope:

UI Testing Automation, Penetration Testing, Compatibility Testing

3. Testing Strategy

Used testing approach goal is to provide ready to present MVP of application that can be used in portfolio. Valuable things that were taken into consideration were time and resources just because that's my project I am working on myself.

Manual Testing

- Happy Path
- Smoke Tests
- Negative Tests
- Exploratory Testing

API Testing

- Endpoint Validation in Postman
- Response / Request Validation
- Error / Invalid handling Testing

ML / OCR Testing

- Entry Data Manipulation
- Image Quality / Position / Environment / Text / Barcode Testing
- Ground Truth Images Testing

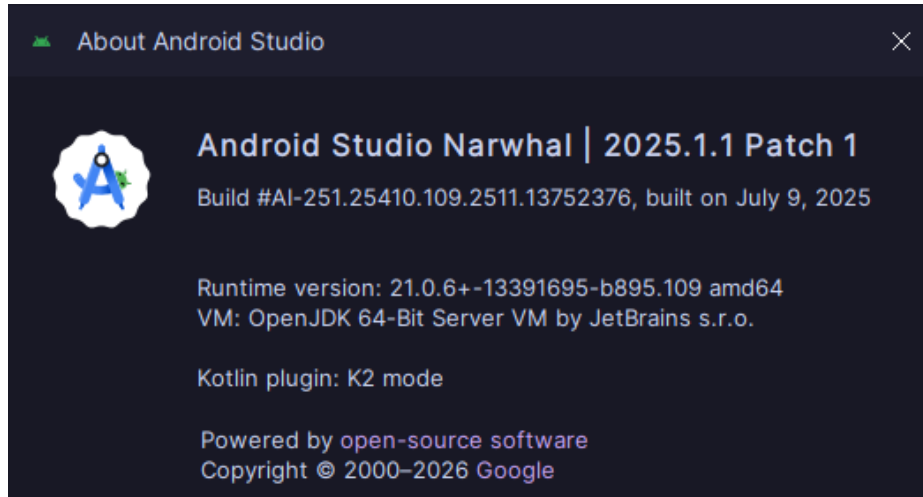
Performance Testing

- Resources (CPU/RAM) Testing
- Handling bigger data sets in UI

4. Test Environment

In order to check all functionalities and flow of application, we made separated testing environment that contains:

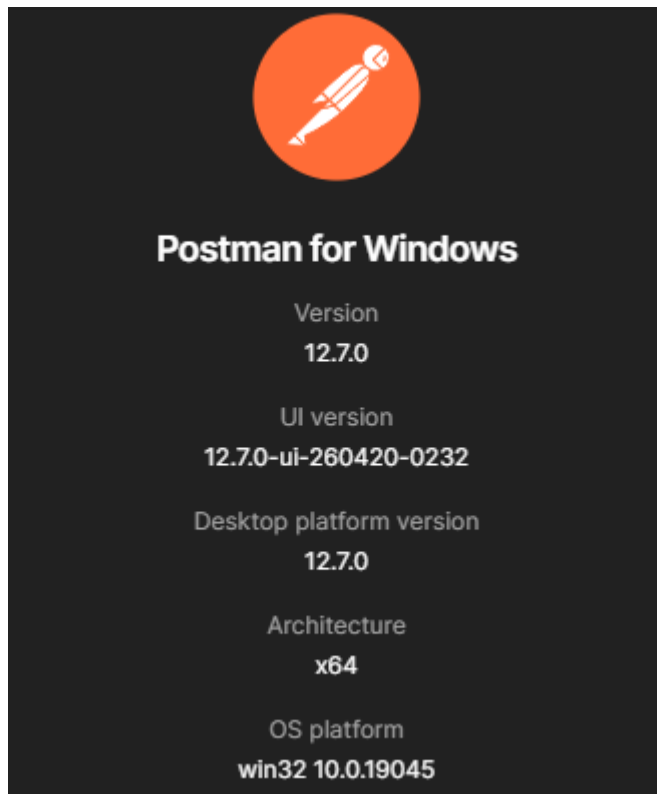
IDE – Android Studio



Emulators / Real Devices –

^	Name	API	Type	
	Medium Phone API 36.0 Android 16.0 ("Baklava") x86_64	36.0	Virtual	▶
	Pixel 4 XL Android 10.0 ("Q") x86	29	Virtual	▶
	Pixel 9 Android 16.0 ("Baklava") x86_64	36.0	Virtual	▶

Postman:



Charles Proxy:



Git:

Ready to use set on newest release version testing branch

Android Studio Profiler

Apache JMeter

5. Entry & Exit Criteria of Testing

Entry (When testing starts?)

When application is building stable version, API connection works and environment is ready to use.

Exit (When testing ends?)

When at least 80% of test cases are done, < 2 critical or high priority and severity bugs detected and still reproduceable, application is in stable condition and every happy path tests are working flawlessly.

6. Deliverables

This section of test plan will present what the result of the testing process is.

Planned deliverables of testing:

- Test Plan
- Test Cases
- Bug Reports
- Postman Collection
- Jira / Xray Stats and reports
- Screenshots